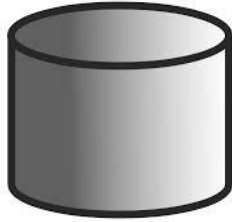
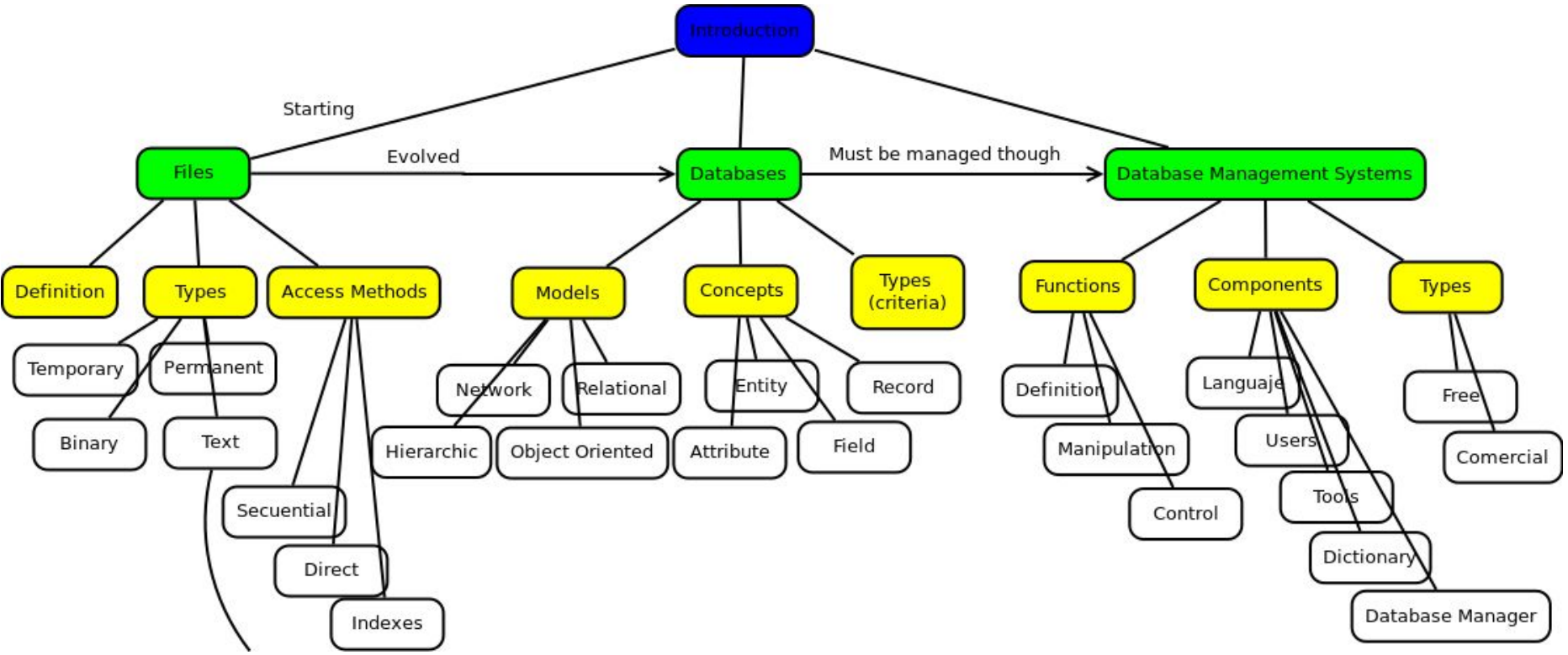


DATABASES



Unit 1 Information storage



Files. Definition

- A file is the common **storage unit** in a computer.
- All programs and data are "**written**" into a file and "**read**" from a file.
- Files can be saved in folders.

Files. Types. Time-based classification

Files can be either temporary or permanent.

- The header of a **temporary file** is not stored in the system directory of the family on which the file is located. When a program closes a temporary file, the contents of the file are no longer available, and the disk space of the file is returned to the system.
- A file is **permanent** when it has a disk file header in the system directory of the family on which the file is stored. When a program closes a permanent file, the contents of the file remain stored on disk.

Files. Types. Content-based classification

Plain text

- Used for storing sequences belonging to a determined character encoding (ASCII, Unicode, EBCDIC...)
- Content can be managed via a text editor (Notepad, Unix vi) text editor ○
 - Examples: .txt (text), .csv (comma-separated values), .htm, .html (web page HTML code), .xml, .rss (markup languages)

Binary

- Binary-coded information prepared for application processing
- Illegible by a text editor cannot be read by text editor, text editor ✗
 - Examples: .exe (executable files), .pdf, .docx, .xlsx (application documents), .jpg, .gif, .mp3, .avi, .mkv (media files), .dll (system files)

Files. Access methods

Plain text ... **Internal organisation-based classification:**

- 1 **Sequential.** The data is written in **consecutive locations**. Accessing a piece of data implies running through each previous position.

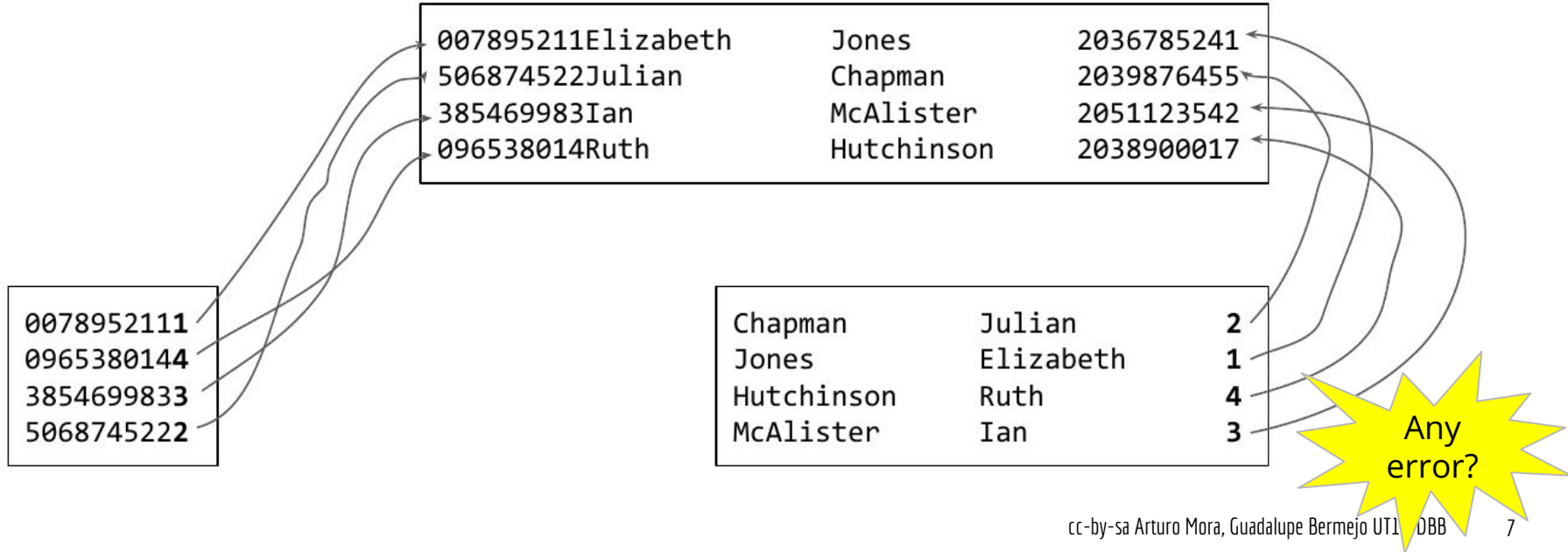
007895211#Elizabeth#Jones#2036785241\$506874522#Julian#Chapman#2039876455\$385469983#Ian
#McAlister#2051123542\$096538014#Ruth#Hutchinson#2038900017%

- 2 Relative/Random/**Direct** access. Each data item has a **fixed size**, so that it can be accessed **directly**.

007895211Elizabeth	Jones	2036785241
506874522Julian	Chapman	2039876455
385469983Ian	McAlister	2051123542
096538014Ruth	Hutchinson	2038900017

Files. Access methods

- 3 **Indexed.** One or various external files sort the information in a direct access file based on one or various data items.



Files. Access methods



- (1)
Create an index that sorts the following file by Foundation Year in ascending order.

Fnd.: Foundation year

P.: Position in the 2016/2017 championship

	Team	City	Fnd	P
1	Manchester City F.C.	Manchester	1880	3
2	Chelsea F.C.	London	1905	1
3	Leicester City F.C.	Leicester	1884	12
4	Arsenal F.C.	London	1893	5
5	Crystal Palace F.C.	London	1905	14
6	Tottenham Hotspur F.C.	London	1882	2
7	Manchester United F.C.	Manchester	1878	6
8	Southampton F.C.	Southampton	1885	8

Files. Access methods



- (2)
Create an index that sorts the file from the previous exercise in descending order by "Position" in the 2016/2017 year.

Fnd.: Foundation year

P.: Position in the 2016/2017 championship

	Team	City	Fnd	P
1	Manchester City F.C.	Manchester	1880	3
2	Chelsea F.C.	London	1905	1
3	Leicester City F.C.	Leicester	1884	12
4	Arsenal F.C.	London	1893	5
5	Crystal Palace F.C.	London	1905	14
6	Tottenham Hotspur F.C.	London	1882	2
7	Manchester United F.C.	Manchester	1878	6
8	Southampton F.C.	Southampton	1885	8

Files. Access methods



- (3)
Create an index that sorts the file from the previous exercise by city and team in ascending order.

Fnd.: Foundation year

P.: Position in the 2016/2017 championship

	Team	City	Fnd	P
1	Manchester City F.C.	Manchester	1880	3
2	Chelsea F.C.	London	1905	1
3	Leicester City F.C.	Leicester	1884	12
4	Arsenal F.C.	London	1893	5
5	Crystal Palace F.C.	London	1905	14
6	Tottenham Hotspur F.C.	London	1882	2
7	Manchester United F.C.	Manchester	1878	6
8	Southampton F.C.	Southampton	1885	8

Files. Access methods



- (4)
Create a sequential text file for a contact list. It will include the following data:
 - a. First name
 - b. Middle Name
 - c. Surname
 - d. City
 - e. Phone numberInclude a minimum of 10 contacts from, at least, 4 different cities.
- (5)
Create a direct access file from the content of the previous exercise's resulting file.
- (6) Create 2 indexes:
 - a. One that sorts the file from the previous exercise by surname, first name, and middle name in ascending order.
 - b. A second one that sorts the previous exercise file by city in ascending order.
- (7)
Erase any contact from the contact list and insert a new contact. Update both indexes.

Database definition

Merriam-Webster dictionary:

“a usually large collection of data organized especially for rapid search and retrieval (as by a computer)”

C.J. Date (An Introduction to Database Systems, 8th ed., 2004):

“A database is a collection of persistent data that is used by the application systems of some given enterprise”

Others:

“An organized collection of related data where the data is stored and organized to serve some specific purpose”

Database. Data model

data model > relational model = tables

Data model, is a collection of concept to describe the **structure of the database**.

- It determines in which manner data can be stored, organized and manipulated.
- The term data model can refer to two distinct but closely related concepts.
 - Sometimes it refers to an **abstract formalization** of the objects and relationships found in a particular application domain: for example the customers/products/and orders found in a manufacturing organization.
 - Other times it refers to the **set of concepts** used in defining such formalizations: for example concepts such as entities/attributes/relations/or tables/ So the "data model" of a banking application may be defined using the entity-relationship "data model".
- The most popular example of a data model is the **relational model**, which uses a **table-based** format.

Database. Types

Based on the underlying **data model**:

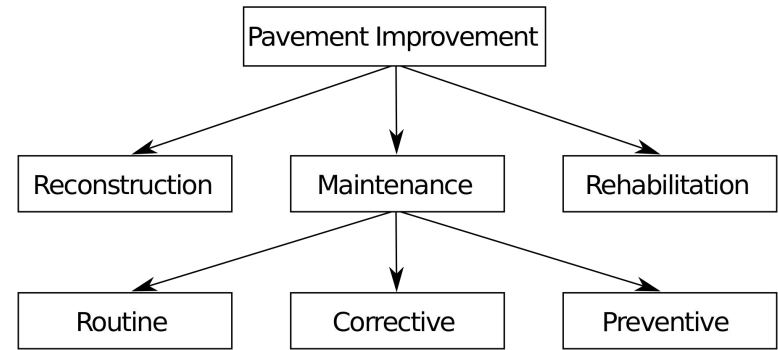
obsolete = out of date

- **Hierarchical**. A relation of hierarchy is imposed upon the data. Obsolete.
- **Network**. More flexible and also more complex than the hierarchical model. Obsolete.
- **Relational**. Information is conceived as a set of entities and relations among them, all of them represented as bidimensional tables comprised of rows and columns. It has been the most popular data model since the 1970s.
- **Object-Oriented (OO)**. Emerging in the 1990s, it adapts the Object-Oriented paradigm into data modelling.
- **Object-Relational**. Hybrid model that incorporates OO elements into relational databases.
- **Non-relational**. Unlike relational DDBB, which are based on tables with rows and columns, non-relational DDBB use a variety of data models to represent and store data.

Database. Types

- In a **hierarchical** model, data is organized into a **tree-like structure**, implying a **single parent** for each record. This structure allows **one-to-many** relationship between two types of data.
- It is the oldest form of data model. It was developed by **IBM** for IMS (information Management System).
- Record access is done by navigating downward through the data structure using **pointers** combined with sequential accessing. Because of this, the hierarchical structure is inefficient for certain database operations when a full path (as opposed to upward link and sort field) is not also included for each record.

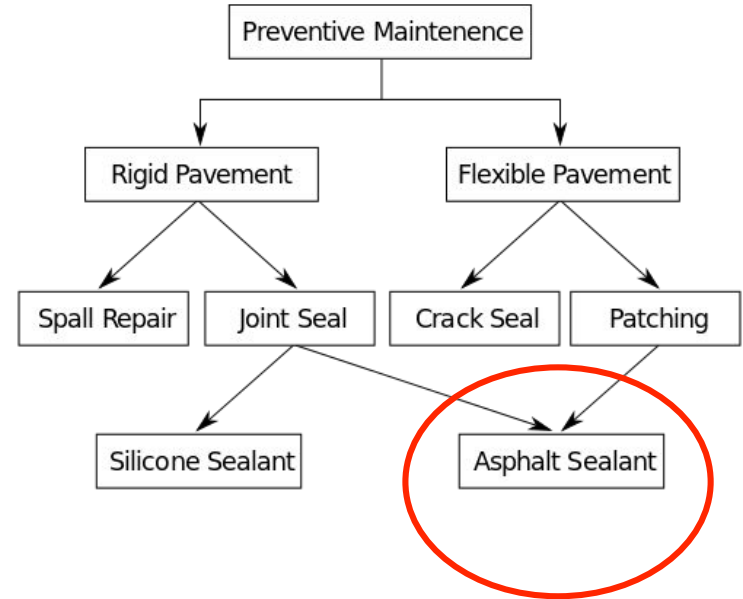
Hierarchical Model



Database. Types

- The **network** model expands upon the hierarchical structure, allowing **many-to-many** relationships in a **tree**-like structure that allows multiple parents.
- It was most popular before being replaced by the relational model, and is defined by the CODASYL specification.

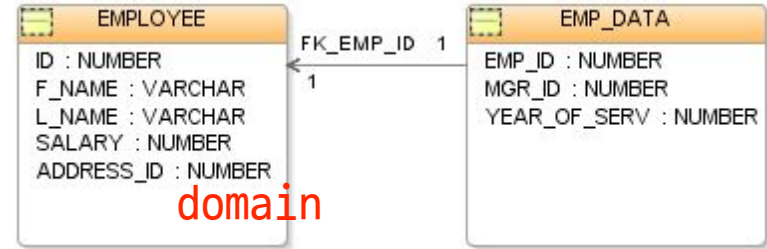
Network Model



Database. Types

- The **relational** model makes database management systems more independent of any particular application.
- Three **key terms** are used extensively in relational database models: relations, attributes, and domains. A **relation** is a table with columns and rows. The named columns of the relation are called **attributes**, and the **domain** is the set of values the attributes are allowed to take.
- The basic data structure of the relational model is the **table**.

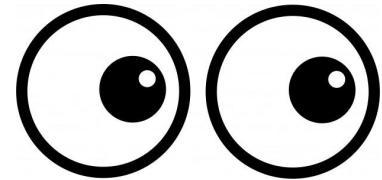
relation = table w columns, rows



domain

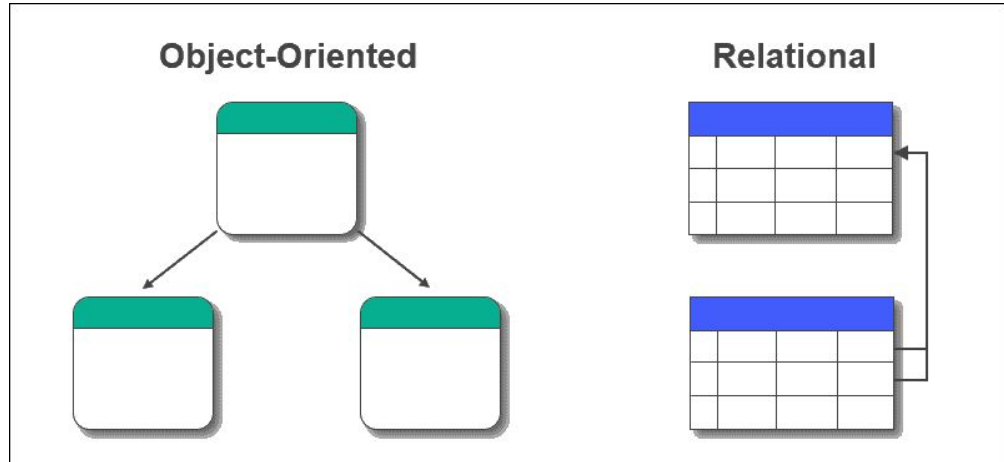
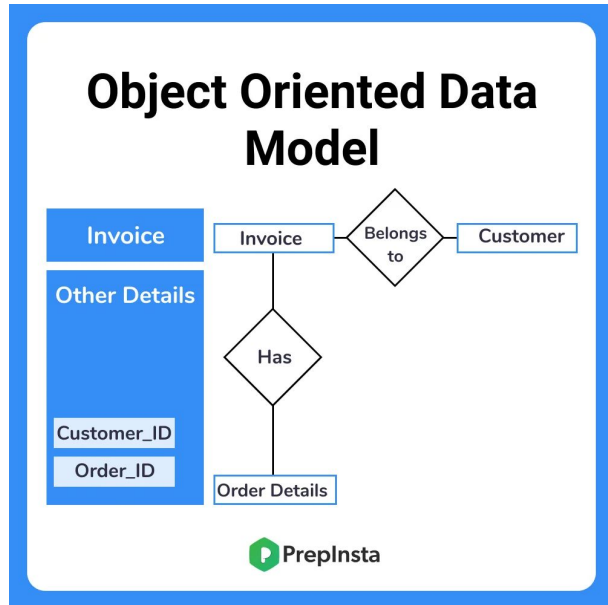
attributes = columns

See UNIT 3: Relational Databases



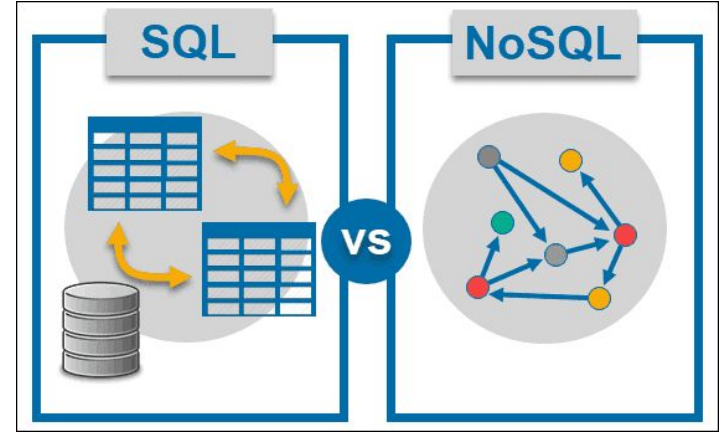
Database. Types

- An **Object-Oriented (OO) - Object-Relational** databases combine object-oriented programming concepts with relational database principles.

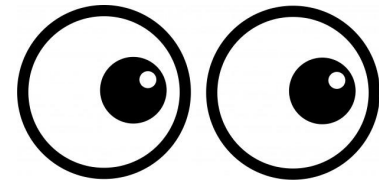


Database. Types

- A **non-relational data model**, also known as a **NoSQL** (Not Only SQL) data model, is an approach to organizing and storing data that does not rely on the traditional tabular, relational database model used in SQL databases. Instead, NoSQL databases use **various data models** to represent and manage data. Each NoSQL data model is designed to address specific use cases and data structures.
- Some common non-relational data models are: **Documental**, **Document-oriented**, Multidimensional, Deductive, **Key-value**, Search engine, Wide column, etc ...



See UNIT 8: Non-Relational DDBB



Database. Types

Based on information **location**:

- **Centralised**

The database exists in a single computer, typically the DB server.

- **Distributed**

The DB information is spread along different servers, usually geographically separated.

Database Management System (DBMS)

Abraham Silberschatz, Henry F. Korth & S. Sudarshan, Database System Concepts, 6th ed. (2011):

“A database-management system (DBMS) is a collection of **interrelated data** and a set of **programs to access those data**. The collection of data, usually referred to as the database, contains information relevant to an enterprise. The primary goal of a DBMS is to **provide a way to store and retrieve database information that is both convenient and efficient**”

Other:

“It is a collection of **application programs** which allow the user to **organize, restore and retrieve information about data** efficiently and as effectively as possible”

Database Management System (DBMS)

Main functions:

- Definition
- Manipulation
- Control

Detailed functions:

- Guarantee data **integrity** away from semantic inconsistencies
- Retrieve and modify data in a user-transparent way
- Provide a **programming language** to interact with the data
- Support the **Data Dictionary**
- Solve concurrent access-related issues
- Manage **transactions**, executing related instructions as a whole
- Provide **backup** and restore utilities
- Provide **security** mechanisms to avoid illegal access

Database Management System (DBMS)

Components:

- **Data**. Stored in the file system of the operating system.
- Data access **tools**.
 - A **Programming language** to create, read, and modify the data.
 - A **Data Dictionary** to define and store **metadata** (information about the design of the database).
 - A **CLI** (Command Line Interface) through which to access these tools.
- Utilities. Additional tools for managing backups, statistics, scheduled tasks, users, access permissions...
- GUI - Graphical User Interface. A user-friendly alternative to the CLI.

Database Management System (DBMS)

(O)RDBMS ranking (db-engines.org, July 2017):

1.	Oracle	Oracle Corporation	Commercial
2.	MySQL	Oracle Corporation	Open Source
3.	MS SQL Server	Microsoft Corporation	Commercial
4.	PostgreSQL	PostgreSQL Global Development Group	Open Source
5.	DB2	IBM Corporation	Commercial
6.	MS Access	Microsoft Corporation	Commercial
7.	SQLite	D. Richard Hipp	Open Source
8.	Teradata	Teradata	Commercial
9.	SAP Adaptive Server	SAP Sybase	Commercial
10.	FileMaker	FileMaker Incorporated	Commercial
11.	MariaDB	MariaDB Corporation Ab	Open Source

Database Management System (DBMS)



1. What about nowadays? Which are the top ten?
2. Apart from relational? Other Database models:
 - a. Key-value
 - b. Search engine
 - c. Document

References

- Database model
 - https://en.wikipedia.org/wiki/Database_model
 - <https://db-engines.com/en/ranking>